Supporting Information

Peak selection matters in Principal Component Analysis: A case study of syntrophic microbes

Running title: Peak selection in syntrophic microbes

Running Authors: Yang et al.

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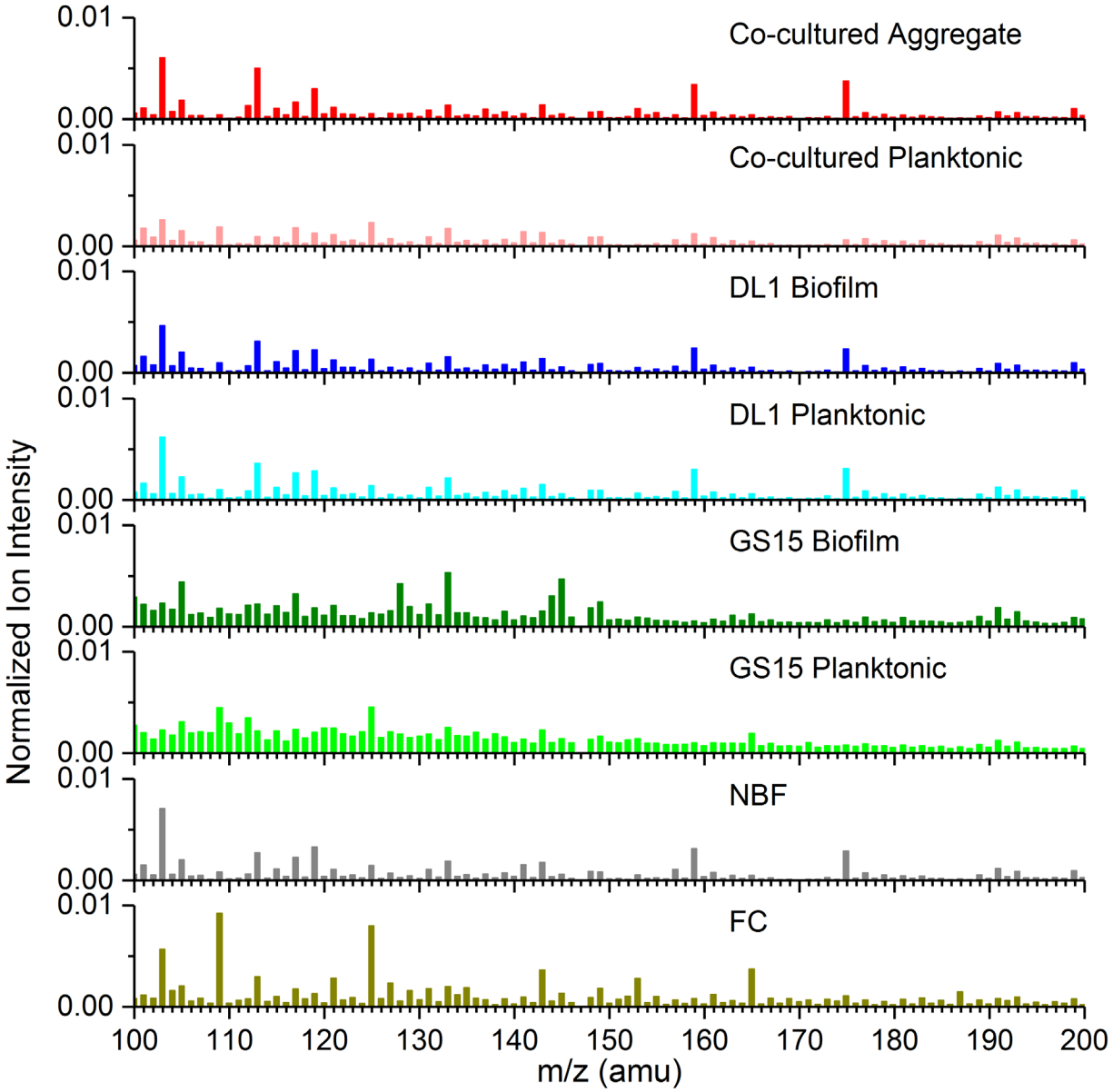


FIG. S1. Comparison of representative positive spectra of eight samples in the range of m/z 100-200.

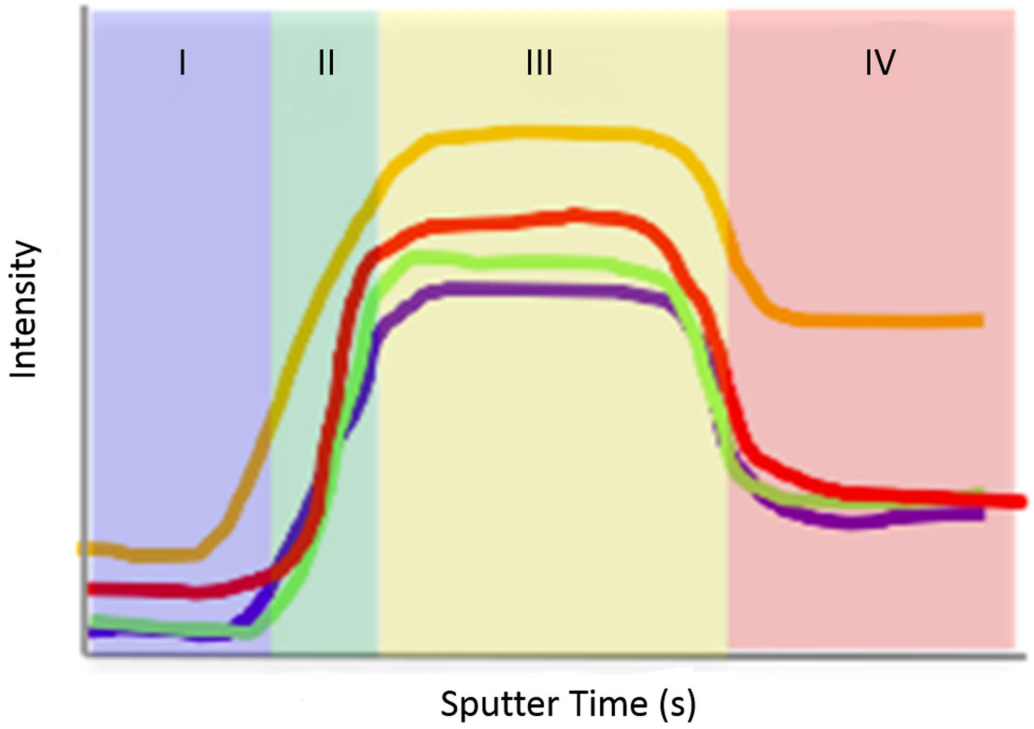


FIG. S2. A schematic showing the *in situ* liquid SIMS process.

Region I corresponds to before SiN punch-through; region II during SiN punch-through; region III acquisition for image reconstruction or high intensity; and region IV acquisition for spectral reconstruction or high resolution.

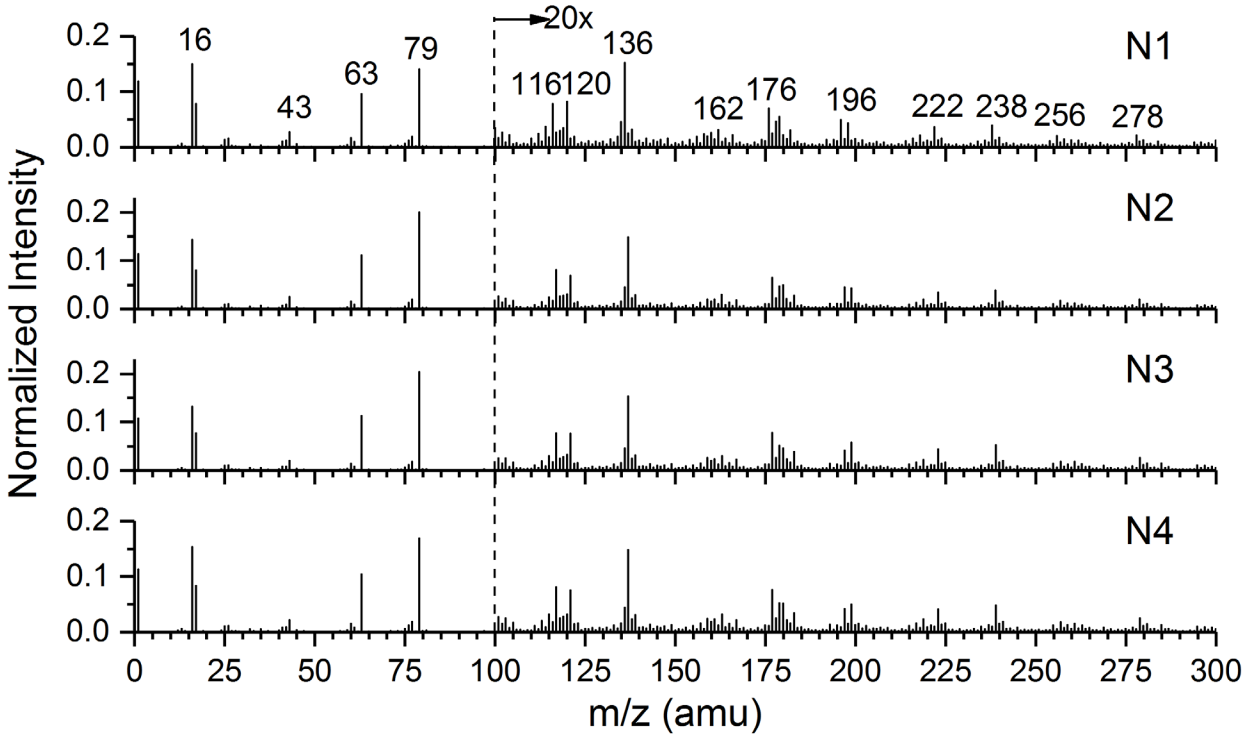


FIG. S3a. Reproducibility of the positive ion spectra of the co-cultured planktonic cells in the range of m/z^+ 0-300.

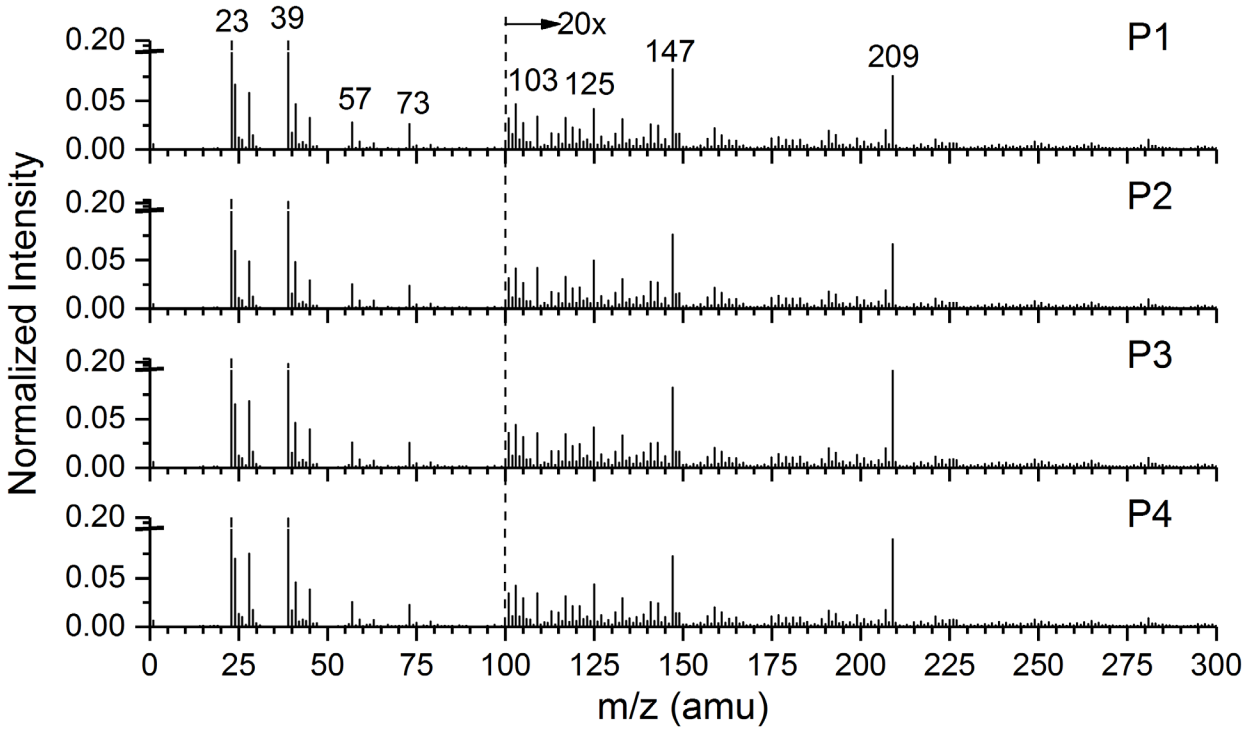


FIG. S3b. Reproducibility of the negative ion spectra of the co-cultured planktonic cells in the range of m/z^+ 0-300.

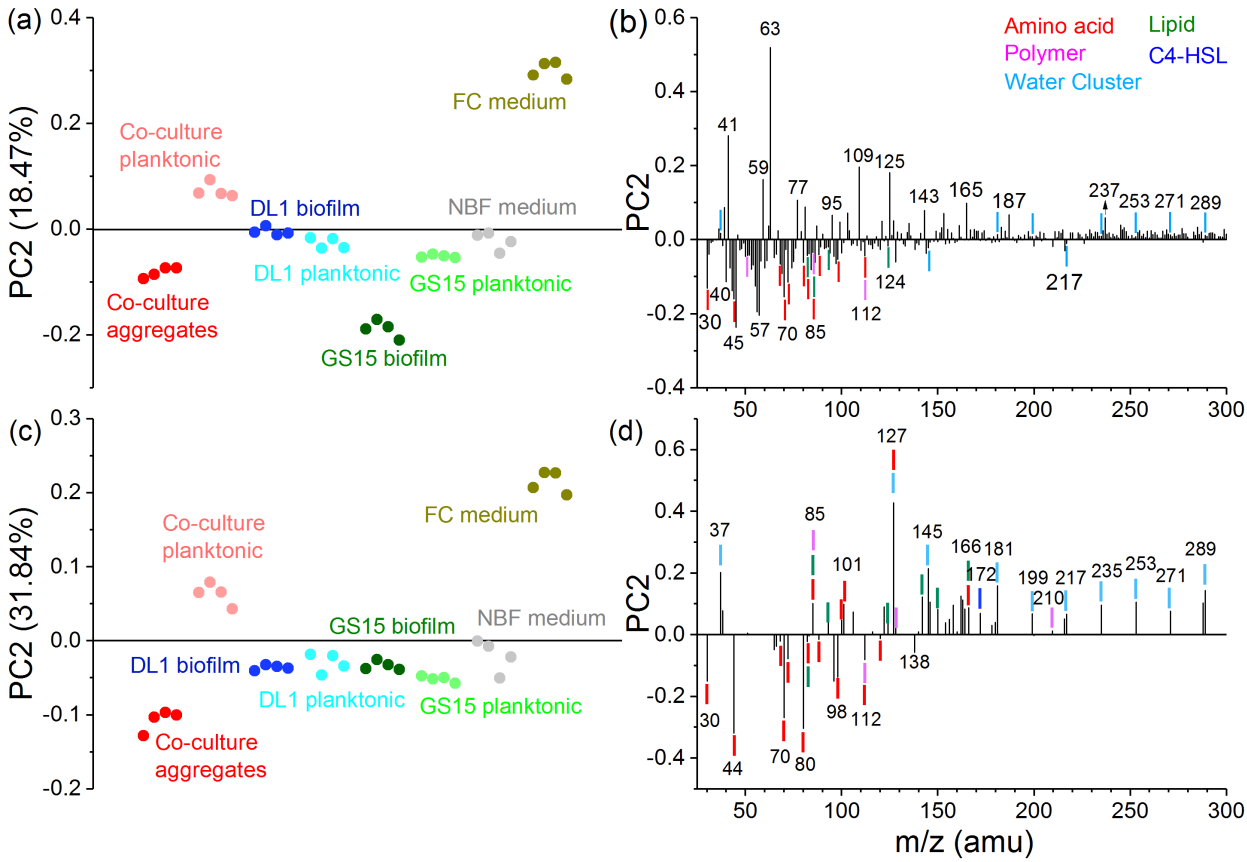


FIG. S4. Comparison of all peak and selected peak spectral PCA in the positive mode. (a) PC2 scores plot of all peak; (b) PC2 loadings plot of all peak; (c) PC2 scores plot of selected peak; and (d) PC2 loadings plot of selected peak.

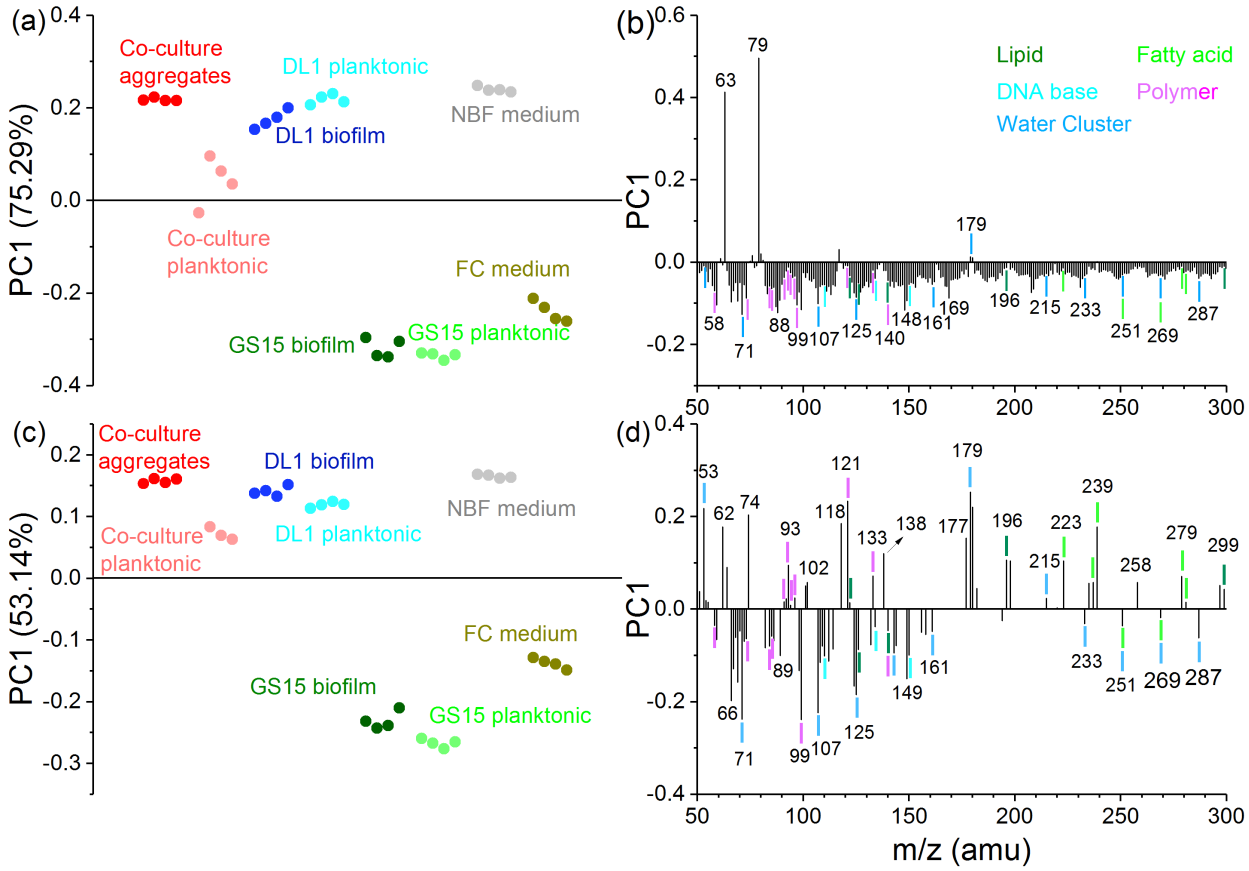


FIG. S5. Comparison of all peak and selected peak spectral PCA in the negative mode. (a) PC1 scores plot of all peak; (b) PC1 loadings plot of all peak; (c) PC1 scores plot of selected peak; and (d) PC1 loadings plot of selected peak.

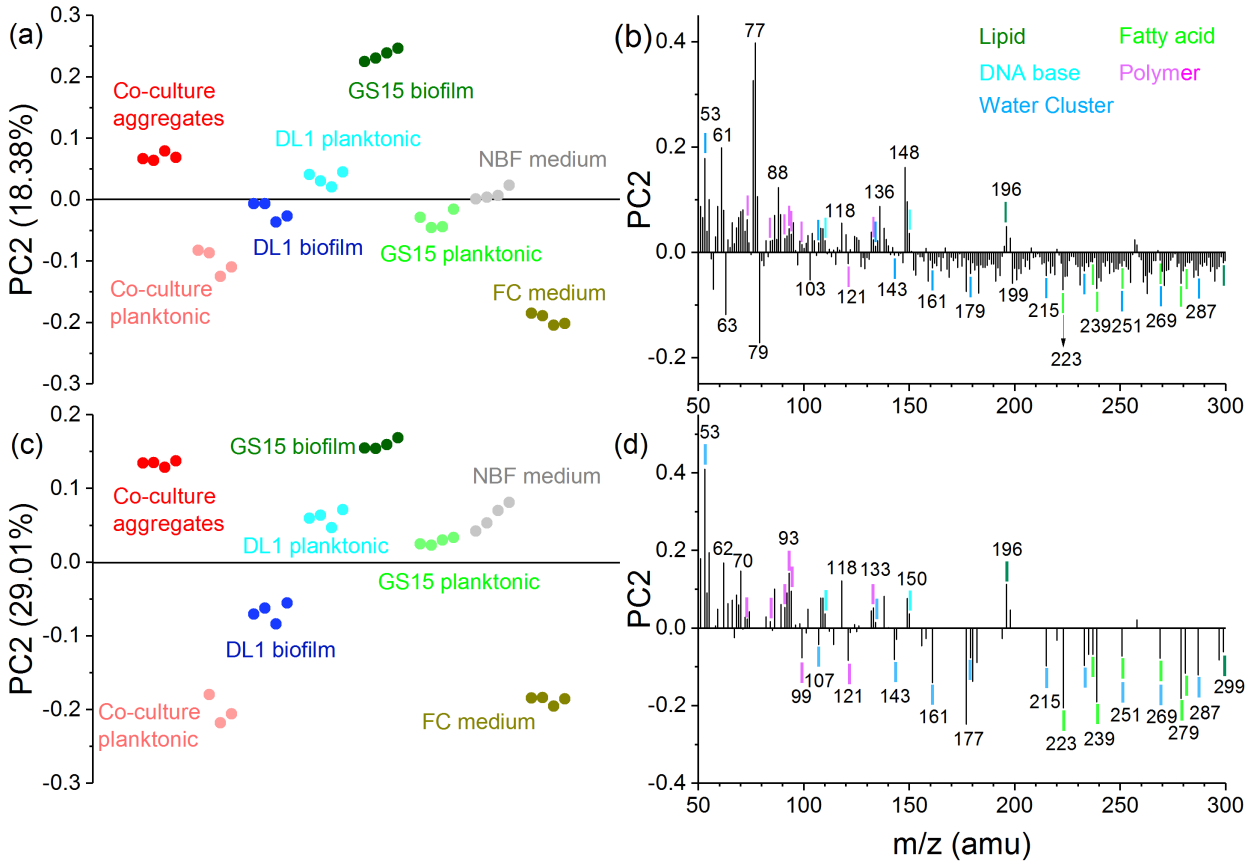


FIG. S6. Comparison of all peak and selected peak spectral PCA in the negative mode. (a) PC2 scores plot of all peak; (b) PC2 loadings plot of all peak; (c) PC2 scores plot of selected peak; and (d) PC2 loadings plot of selected peak.

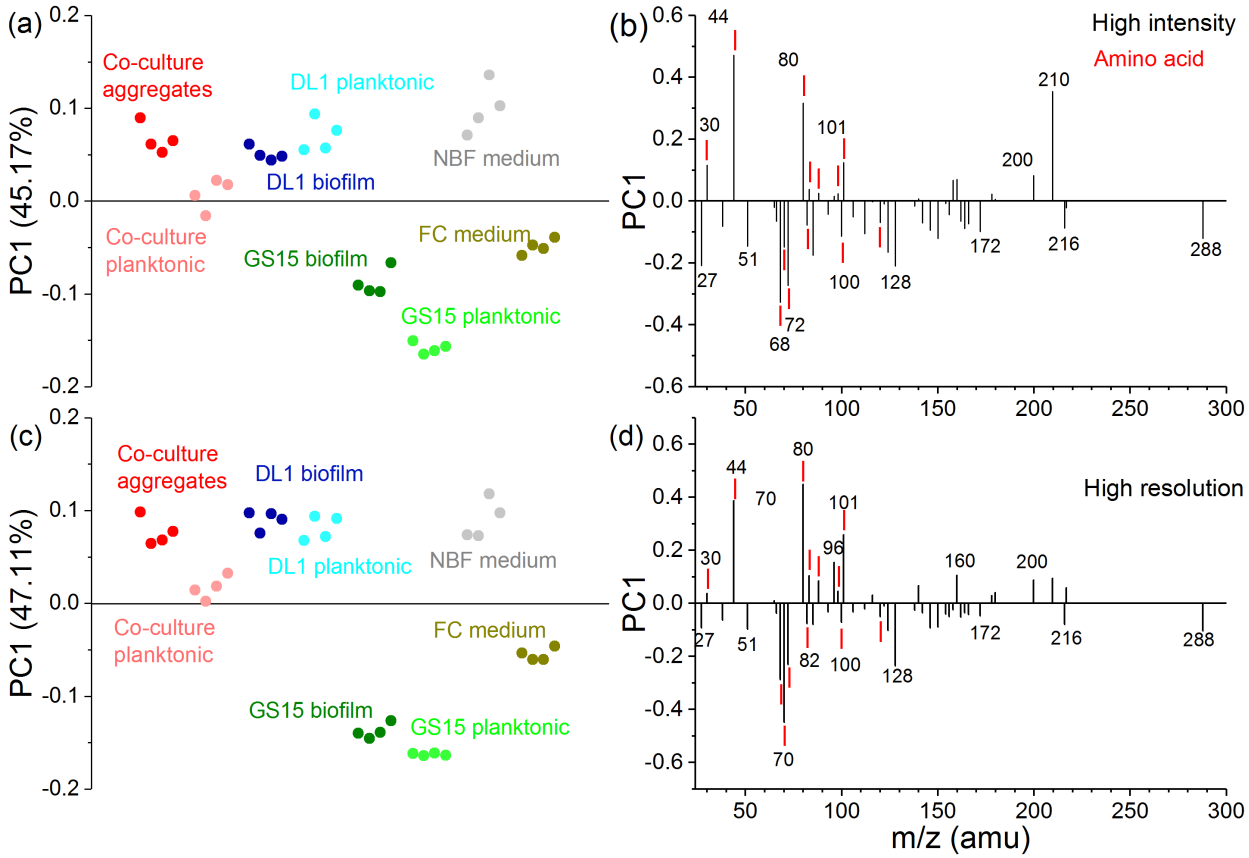


FIG. S7. Comparison of selected peak spectral PCA between high intensity and high resolution data in the positive mode. (a) PC1 scores plot of high intensity data; (b) PC1 loadings plot of high intensity data; (c) PC1 scores plot of high resolution data; and (d) PC1 loadings plot of high resolution data.

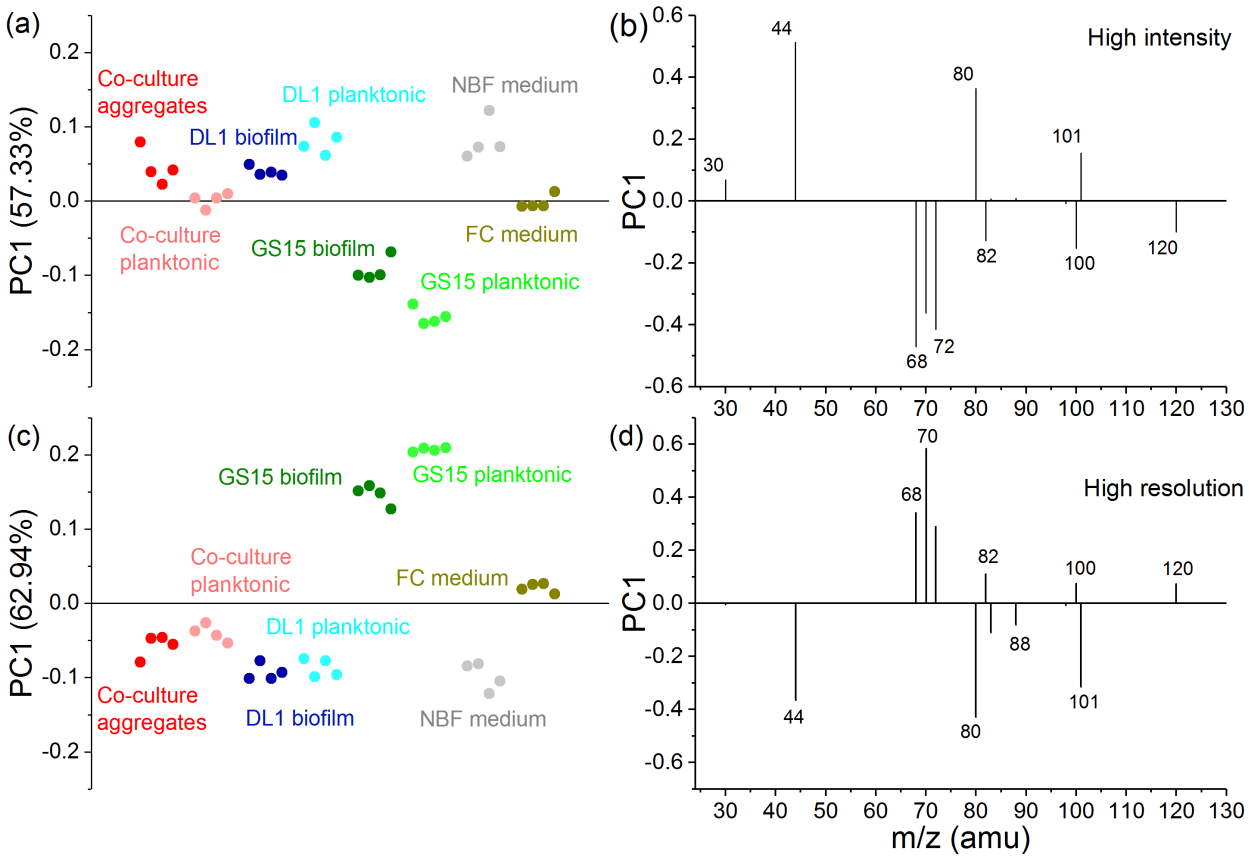


FIG. S8. Comparison of amino acid peak spectral PCA between high intensity and high resolution data in the positive mode. (a) PC1 scores plot of high intensity data; (b) PC1 loadings plot of high intensity data; (c) PC1 scores plot of high resolution data; and (d) PC1 loadings plot of high resolution data.

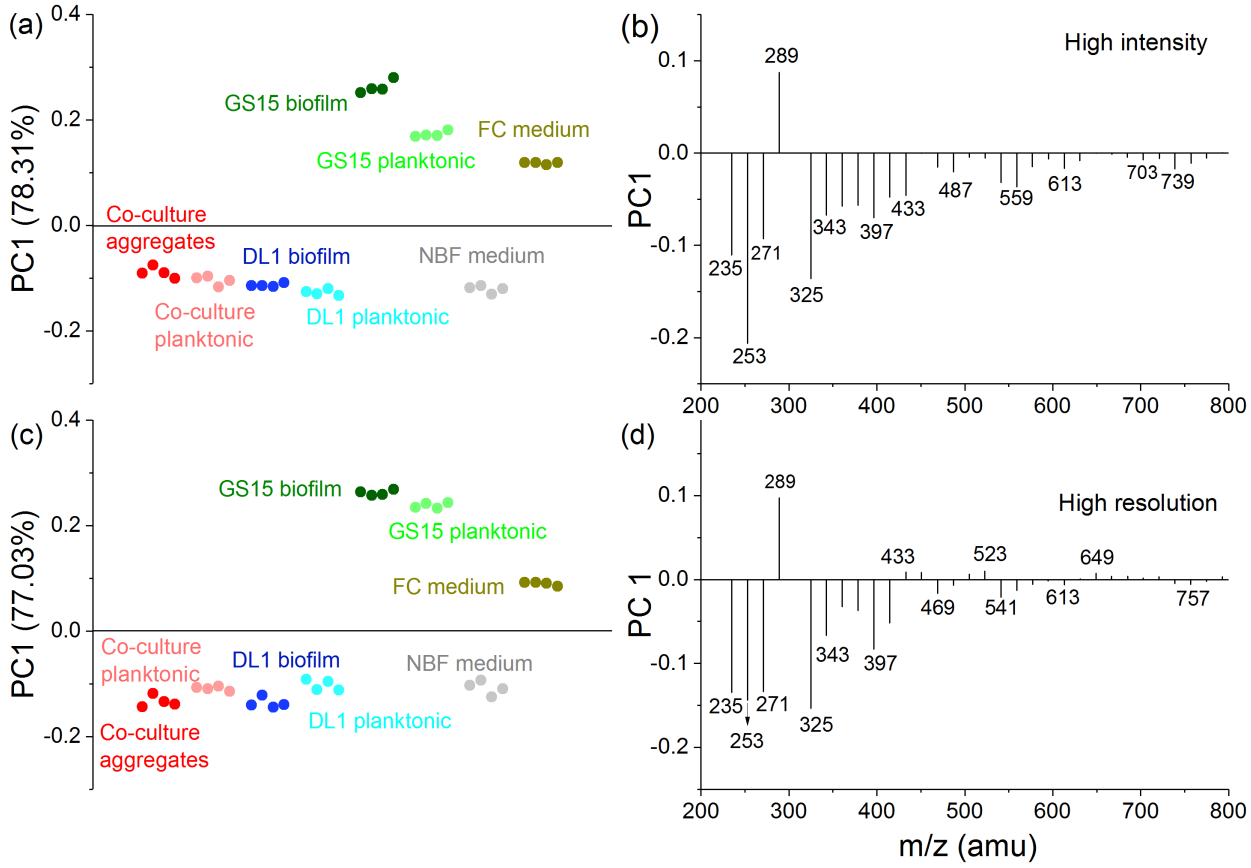


FIG. S9. Comparison of water cluster peak spectral PCA between high intensity and high resolution data in the positive mode. (a) PC1 scores plot of high intensity data; (b) PC1 loadings plot of high intensity data; (c) PC1 scores plot of high resolution data; and (d) PC1 loadings plot of high resolution data.

Supplemental Tables

Table S1 and Table S2 show possible peak assignments in the positive mode and negative mode, respectively.

Table S1. Possible peak assignment in the positive mode

Unit mass (m/z)	m/z^+ obs.	m/z^+ the.	Suggested Formula	Description	References
19	19.007	19.018	$(H_2O)\ H^+$	Water cluster	1, 2
30	30.002	30.034	CH_4N^+	Gly	3-6
37	37.024	37.029	$(H_2O)_2\ H^+$	Water cluster	1, 2
39	38.972	38.964	K^+	medium	7
41	40.998	41.039	$C_3H_5^+$	Polysaccharides	8
44	43.993	44.050	$C_2H_6N^+$	Ala	3-5
51	50.998	51.024	$C_4H_3^+$	$C_xH_y^+$ Chains of polymer	9
56	55.938	55.935	Fe^{2+}	medium	6
57	56.972	56.965	$CaOH^+$	medium	2
63	62.984	62.982	Na_2OH^+	medium	6
68	67.988	68.050	$C_4H_6N^+$	Pro/Lys	3, 5, 6
70	70.003	70.029	$C_3H_4NO^+$	Asn	3, 4
72	71.980	72.081	$C_4H_{10}N^+$	Val	3, 4, 10
80	79.994	80.050	$C_5H_6N^+$	Pro	10, 11
82	81.996	82.053/ 82.066	$C_4H_6N_2^+$/ $C_5H_8N^+$	His/Phosphatidylcholine	12-14
83	82.994	83.050	$C_5H_7O^+$	Val/Glu	3, 4
85	84.994	85.102/85.065/85.065/85.077/85.040	$C_6H_{13}^+$/ $C_5H_9O^+$/ $C_4H_5O_2^+$/$C_4H_9N_2^+$/ $C_3H_5N_2O^+$	Phosphatidylcholine/ Polymer/ Gly	8, 14-16
88	87.994	88.040	$C_3H_6NO_2^+$	Asn/Asp	3-5
91	90.976	91.061	$(H_2O)_5\ H^+$	Water cluster	1, 2
93	92.987	93.070	$C_7H_9^+$	phosphatidylcholine	14
95	95.016	95.086	$C_7H_{11}^+$	sterol lipid	17
98	97.997	98.024	$C_4H_4NO_2^+$	Asn	3, 4
100	99.973	100.087	$C_4H_{10}N_3^+$	Arg	3, 6
101	100.976	101.095	$C_4H_{11}N_3^+$	Arg	3
112	111.943	112.087/112.076	$C_5H_{10}N_3^+$/$C_6H_{10}NO^+$	Arg/polymer	12, 13, 16, 18
120	119.974	120.081	$C_8H_{10}N^+$	Phe	3-5, 19
124	123.984	124.016	$C_2H_7NPO_3^+$	GPEtn(Lipids)	20
128	127.951	128.063	$C_{10}H_8^+$	polymer	9
142	141.945	142.027	$C_2H_9NPO_4^+$	DPPE headgroup fragment	20, 21
145	144.950	145.092	$(H_2O)_8\ H^+$	Water cluster	1, 2
150	149.955	150.068	$C_5H_{13}NPO_2^+$	Phosphatidylcholine-fragments	14
159	158.921	159.029	$C_6H_7O_5^+$	polymer	22
163	162.948	163.103	$(H_2O)_9H^+$	Water cluster	1, 2
166	165.955	166.063/ 166.087	$C_5H_{13}NPO_3^+$/$C_9H_{12}NO_2^+$	Phosphatidylcholine or Phenylalanine	14, 16, 20
172	171.932	172.097	$C_8H_{14}NO_3^+$	N-butyryl-L-Homoserine lactone	2, 6
175	174.908	175.063	$C_{10}H_9NO_2^+$	quinolone signal	23-25
181	180.8960	181.113	$(H_2O)_{10}\ H^+$	Water cluster	1, 2
199	198.880	199.124	$(H_2O)_{11}\ H^+$	Water cluster	1, 2
217	216.863	217.135	$(H_2O)_{12}\ H^+$	Water cluster	1, 2
235	234.862	235.145	$(H_2O)_{13}H^+$	Water cluster	1, 2
271	270.816	271.166	$(H_2O)_{15}H^+$	Water cluster	1, 2
289	288.804	289.177	$(H_2O)_{16}H^+$	Water cluster	1, 2
325	324.786	325.198	$(H_2O)_{18}H^+$	Water cluster	1, 2

343	342.724	343.209	$(H_2O)_{19}H^+$	Water cluster	[1, 2]
379	378.667	379.230	$(H_2O)_{21}H^+$	Water cluster	[1, 2]
415	414.644	415.251	$(H_2O)_{23}H^+$	Water cluster	[1, 2]
451	450.705	451.272	$(H_2O)_{25}H^+$	Water cluster	[1, 2]
487	486.927	487.293	$(H_2O)_{27}H^+$	Water cluster	[1, 2]
505	504.896	505.304	$(H_2O)_{28}H^+$	Water cluster	[1, 2]
523	522.875	523.314	$(H_2O)_{29}H^+$	Water cluster	[1, 2]
541	540.946	541.325	$(H_2O)_{30}H^+$	Water cluster	[1, 2]
577	576.949	577.346	$(H_2O)_{32}H^+$	Water cluster	[1, 2]
595	594.927	595.356	$(H_2O)_{33}H^+$	Water cluster	[1, 2]
631	631.034	631.378	$(H_2O)_{35}H^+$	Water cluster	[1, 2]
667	666.941	667.399	$(H_2O)_{37}H^+$	Water cluster	[1, 2]
703	702.979	703.420	$(H_2O)_{39}H^+$	Water cluster	[1, 2]
757	756.995	757.452	$(H_2O)_{42}H^+$	Water cluster	[1, 2]
793	793.002	793.473	$(H_2O)_{44}H^+$	Water cluster	[1, 2]

Note: m/z_{obs}: observed mass to charge ratio; m/z_{the}: theoretical mass to charge ratio.

Table S2. Possible peak assignment in the negative mode

Unit mass (m/z^-)	$m/z^-_{obs.}$	m/z^-_{the}	Suggested Formula	Possible Description	References
53	53.004	53.024	$(H_2O)_2OH^-$	Water cluster	1, 2
58	58.001	58.005	$C_2H_2O_2^-$	polymer	26
63	62.961	62.964	PO_2^-	medium	2
71	71.010	71.034	$(H_2O)3OH^-$	Water cluster	1, 2
73	72.986	73.008	C_6H^-	polymer	15
79	78.958	78.959	PO_3^-	medium	2
84	83.990	84.045	$C_4H_6NO^-$	polymer	18
85	84.997	85.029	$C_4H_5O_2^-$	polymer	26
91	90.968	91.018	$C_6H_3O^-$	polymer	26
93	92.997	93.034	$C_6H_5O^-$	Benzene (Polymers)	27
94	93.983	94.042	$C_6H_6O^-$	Benzene (Polymers)	27
96	95.970	96.009	$C_4H_2NO_2^-$	Unreacted maleimide ring /maleimide-thiol bond (Polymers)	4
99	98.988	99.081	$C_6H_{11}O^-$	Large saturated ring (Polymers)	27
107	106.954	107.056	$(H_2O)_5OH^-$	Water cluster	1, 2
110	109.966	110.035	$C_4H_4N_3O^-$	Cyt-H DNA base	28, 29
121	120.949	121.029	$C_7H_5O_2^-$	Unreacted acrylate group (Polymers)	27
122	121.951	122.001	$C_2H_5NO_3P^-$	Phosphatidylethanolamine (Lipid)	30
125	124.980	125.066	$(H_2O)_6OH^-$	Water cluster	1, 2
126	125.970	126.032	$C_2H_9NPO_3^-$	2-aminoethylphosphono lipids	17
133	132.949	133.065	$C_9H_9O^-$	polymer	15
134	133.939	134.047	$C_5H_4N_5^-$	Ade-H DNA base	31
140	139.968	140.011/139.993	$C_2H_7NPO_4^-$ /$C_6H_4SO_2$-	Phosphatidylethanol-Amine or Polymers	18, 30, 32, 33
143	142.948	143.077	$(H_2O)_7OH^-$	Water cluster	1, 2
150	149.928	150.042	$C_5H_4N_5O^-$	Gua-H DNA base	28
161	160.901	161.087	$(H_2O)_8OH^-$	Water cluster	1, 2
179	178.898	179.098	$(H_2O)_9OH^-$	Water cluster	1, 2
196	195.904	196.037	$C_5H_{11}NPO_5^-$	Phosphatidylethanol-amine (Lipids)	33
215	214.856	215.119	$(H_2O)_{11}OH^-$	Water cluster	1, 2
223	222.847	223.170	$C_{14}H_{23}O_2^-$	Phosphoinositol or phosphatidylinositol (Fatty acid)	34-36
233	232.834	233.130	$(H_2O)_{12}OH^-$	Water cluster	1, 2
237	236.863	237.222	$C_{16}H_{29}O^-$	Myristic acid	2
239	238.854	239.201	$C_{15}H_{27}O_2^-$	Aliphatic chains of Fatty acid	2
251	250.843	251.140/251.201	$(H_2O)_{13}OH^-$ /$C_{16}H_{27}O_2^-$	Water cluster or Fatty acid	1, 2, 34, 37
269	268.846	269.151/269.248	$(H_2O)_{14}OH^-$/ $C_{17}H_{33}O_2^-$	Water cluster or Fatty acid	1, 2, 34, 37
279	278.784	279.232	$C_{18}H_{31}O_2^-$	Linoleic acid	2
281	280.802	281.248	$C_{18}H_{33}O_2^-$	Oleic acid	30, 33, 34
287	286.801	287.161	$(H_2O)_{15}OH^-$	Water cluster	1, 2

299	298.794	299.053	$C_9H_{16}PO_9^-$	Phosphoinositol (lipids)	[38]

Note: m/z_{obs}: observed mass to charge ratio; m/z_{the}: theoretical mass to charge ratio.

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